1. Clinical Question & Core Hypothesis

Clinical Question

From the Ontologist synthesis, TIL/imaging, serum C-markers, and proliferation (Ki67) each correlate with outcome under PI3K inhibition, but they sometimes give discordant signals — e.g., immune-infiltrated tumors that nonetheless fail to benefit from alpelisib. Why does immune infiltration not uniformly translate into therapy sensitivity? 2) Tier 1 paths used: I built on CORE PATH 1 (Biomarker:C-markers) because serial serum signatures can report tumor metabolic/PI3K activity and pharmacodynamic state; and CORE PATH 3 (Biomarker:Ki67 expression) because tumor proliferation plausibly modulates immune efficacy and drug sensitivity. I substituted an imaging feature allowable in the dataset (TIL density) to capture the immune microenvironment that the Ontologist's SUVmax path implied but which is not in the permitted imaging list. 3) Tier 2 support: PD-L1, ROR score, and pan-immune inflammation value provide orthogonal immune/inflammatory and risk-context that could modulate the TIL–proliferation–serum marker interaction (e.g., PD-L1/exhaustion status or systemic inflammation may amplify or blunt TIL functional capacity). 4) Why therapy_response endpoint: The primary translational question is who will derive clinical benefit from PI3Kα inhibition (alpelisib), so a therapy_response proxy (response to PI3K inhibition) is the most actionable, mechanistic endpoint for trial stratification and for linking serial serum/imaging biomarkers to immediate clinical decision-making. 5) What I excluded and why: I deliberately excluded SUVmax as the core imaging variable because the allowed imaging features list restricts us to TIL/TLS/TSR; I excluded EndoPAT/CA19-9/Endothelial markers (Tier 3) from the core claim to keep the hypothesis sharply focused on immune–proliferation–metabolic serum interactions; Tier 2 items were relegated to secondary_modifiers to preserve the Focusing Rule (max 2 anchors)?

Working Hypothesis

Expanded Hypothesis: In HR+/HER2− breast cancer, the predictive value of intratumoral TIL density for response to PI3Kα inhibition depends on tumor proliferation and systemic metabolic signaling. Specifically, TIL-high tumors will exhibit greater therapy_response to PI3Kα inhibition when Ki67 expression is low and the serum C-marker signature indicates limited PI3K-driven metabolic activity; if Ki67 is high or C-marker signals reflect active PI3K-driven metabolism, the advantage conferred by high TIL density is reduced or absent. Therefore, a three-factor classifier integrating TIL density (high/low), Ki67 (low/high), and C-marker serum profile (low/high) will more accurately predict therapy_response than any single biomarker. This framework implies mechanistic crosstalk where PI3K signaling modulates tumor metabolism and immune cell efficacy, and where proliferation rate shapes immune contexture responses to PI3Kα inhibitors.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	Biomarker:C-markers, Biomarker:Ki67 expression
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

Biomarker:C-markers: Biomarker:C-markers) because serial serum signatures can report tumor metabolic/PI3K activity and pharmacodynamic state; and CORE PATH 3 (Biomarker:Ki67 expression) because tumor proliferation plausibly modulates immune efficacy and drug sensitivity.

Biomarker:Ki67 expression: Biomarker:Ki67 expression) because tumor proliferation plausibly modulates immune efficacy and drug sensitivity.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Biomarker:programmed cell death 1 ligand 1	Supporting modifier from knowledge graph
Secondary	Biomarker:ROR score	Supporting modifier from knowledge graph
Secondary	Biomarker:pan-immune inflammation value	Supporting modifier from knowledge graph
Secondary	Biomarker:IHC expressions of ER, PR, or HER2	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective analyses in HR+/HER2− cohorts treated with PI3Kα inhibitors to evaluate three-way interactions among TIL density, Ki67, and C-marker signature on therapy_response and PFS;

Prediction 2: prospective stratified cohorts or biomarker-driven trial-arm design to validate the classifier;

Prediction 3: mechanistic preclinical studies using tumor organoids or co-culture systems with T cells to assess how PI3Kα inhibition affects TIL function under varying Ki67 and metabolic states;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Biomarker:programmed cell death 1 ligand 1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Biomarker:ROR score: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.